Excess entropy and divalent-ion mobility in multicomponent charged LJ mixtures

HNC screening · replicated MD · thermodynamic integration, pair entropy and 2PT · two divalent concentrations

+0.30 kB

excess entropy gained by HNC-selected solvent designs

+18–20%

divalent-ion self-diffusion, both signs

0%

effect of adding labels without changing interactions

Design: label count vs interaction diversity

1

HNC screen

35 integer-charge, composition and dispersion candidates at $T^* = 2$, $\rho^* = 0.7$; 8 selected

2

Replicated MD

8 mixtures × 3 matched replicas, N = 4000, x(±2) = 0.03; probes matched across mixtures

3

Entropy and transport

TI over 19 densities (fixed LJ cutoff, Debye–Hückel endpoint); MSD in one common COM frame

Code	Mixture (all with 3% of each ±2 probe)	Role
B3	94% LJ solvent	Baseline
B5	B3 + 5% of each ±1 ion	Salt addition
U4	64% solvent + 30% weak solvent, ε = 0.25	Reference
L5	U4 with 2 identical weak-solvent labels	Label null
M5	U4 split into ε = 0.0625 / 0.5625	Moderate split
H5	U4 split into ε = 0.01 / 0.81	Strong split
U6 / H7	U4 / H5 plus 5% of each ±1 ion	With salt

U4 → M5 → H5 keep charged composition, ionic strength and the mean bare LJ amplitude $\Sigma_{ij} x_i x_j \varepsilon_{ij} = 0.7225$ fixed

Seven kinds of particles, one size

Solvent A
q = 0, ε = 1

all mixtures, 54–94%

Weak solvent B
q = 0, ε = 0.25

U4, L5, U6: 30%

Split solvent B_1 / B_2
q = 0, ε = 0.01 / 0.81 (H)
or 0.0625 / 0.5625 (M)

M5, H5, H7: 15% each

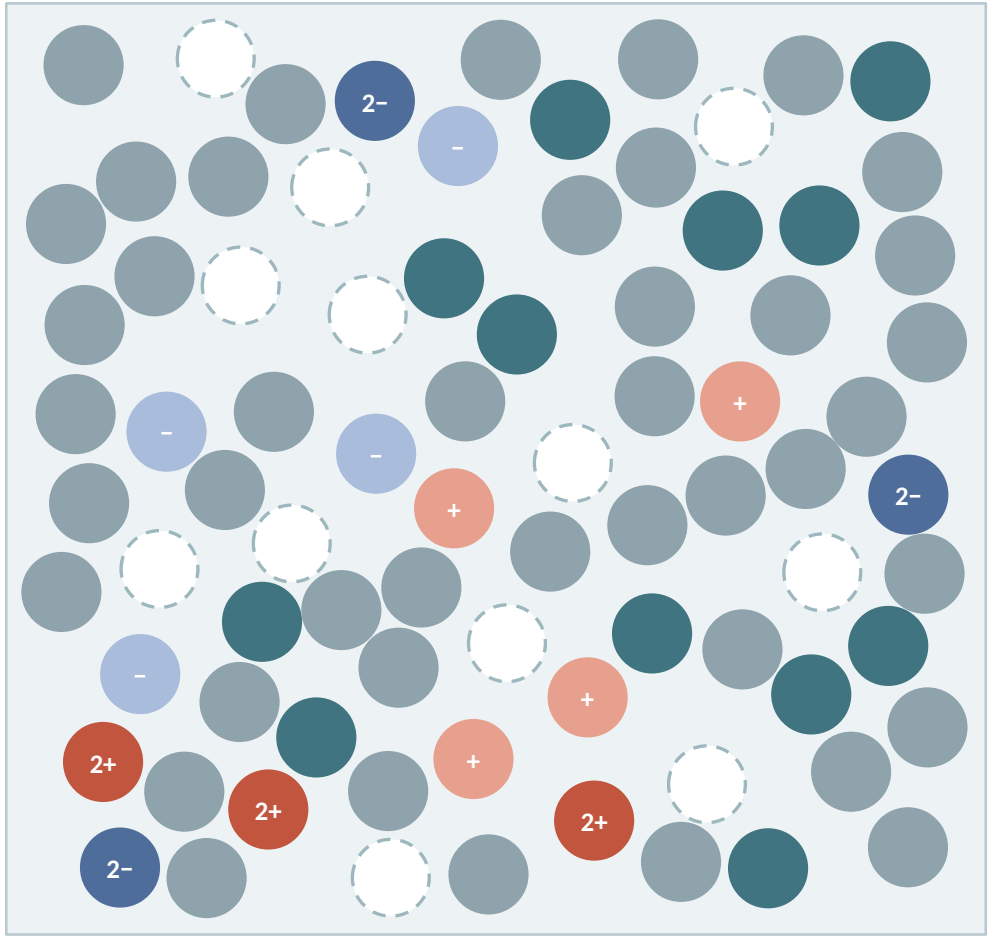
Divalent probe ions
q = ±2, ε = 1

all mixtures: 3% each

Monovalent background
q = ±1, ε = 1

B5, U6, H7: 5% each

All particles: σ = 1, m = 1; $\varepsilon_{ij} = \sqrt{(\varepsilon_i \varepsilon_j)}$; Coulomb with Bjerrum length 0.5σ at $T^* = 2$



Cartoon of H7 (2D sketch with roughly its proportions; not simulation data)

HNC for charged mixtures in four steps

1 **Pair structure** $g_{ij}(r),\ h_{ij} = g_{ij} - 1$

Probability of finding a j particle at distance r from an i particle, relative to a random mixture

2 **Ornstein–Zernike equation** $h_{ij} = c_{ij} + \Sigma_k \rho_k c_{ik} \otimes h_{kj}$

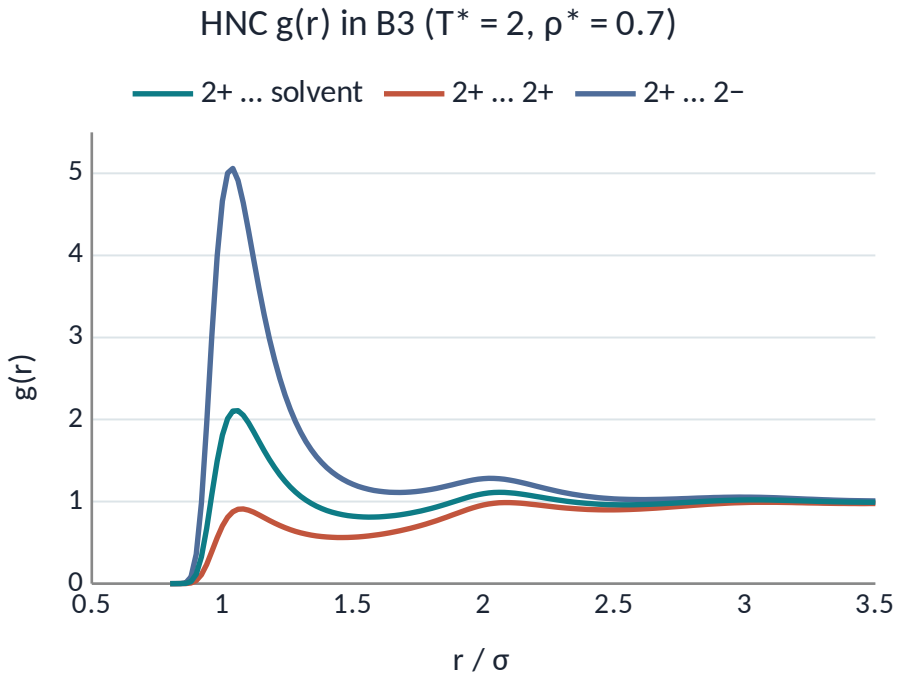
Total correlation = direct part + indirect part relayed through every species k (a p × p matrix equation in k space)

3 **HNC closure (Morita–Hiroike)** $g_{ij} = \exp(-\beta u_{ij} + h_{ij} - c_{ij})$

Second relation that closes the system; it neglects the bridge diagrams and is accurate for ions

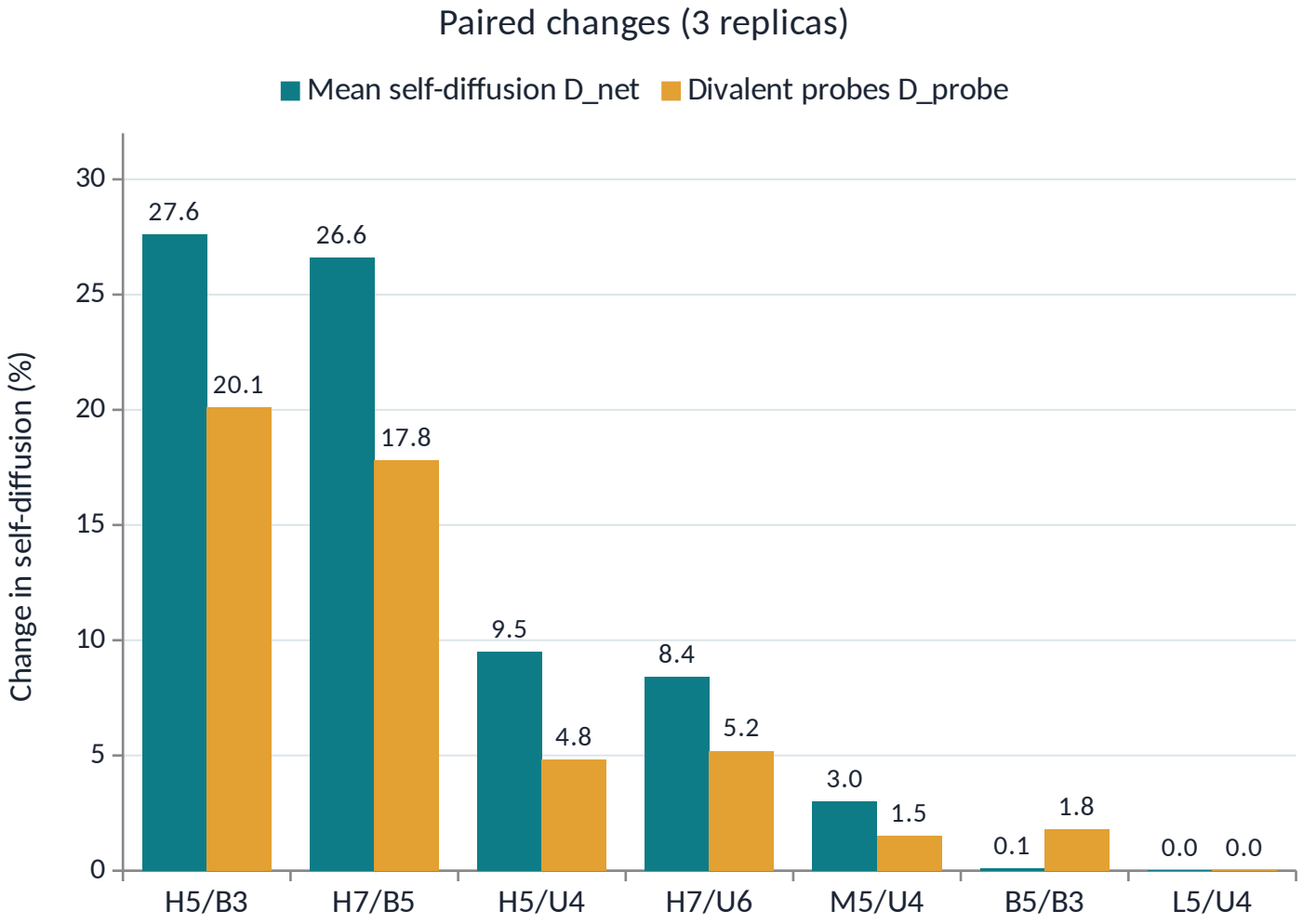
4 **Charges (Allnatt; Lomba–Høye)** $c_{ij} \to -\lambda_B q_i q_j / r$

Coulomb tails decay too slowly to Fourier transform: subtract the screened (Debye) tail analytically and solve for short-ranged parts



Why it is useful here: in HNC the free energy follows in closed form from g and c, so $s_{ex} = u_{ex} - a_{ex}$ costs one solution per mixture, enough to screen dozens of compositions

Entropy speeds up ions, less than the solvent



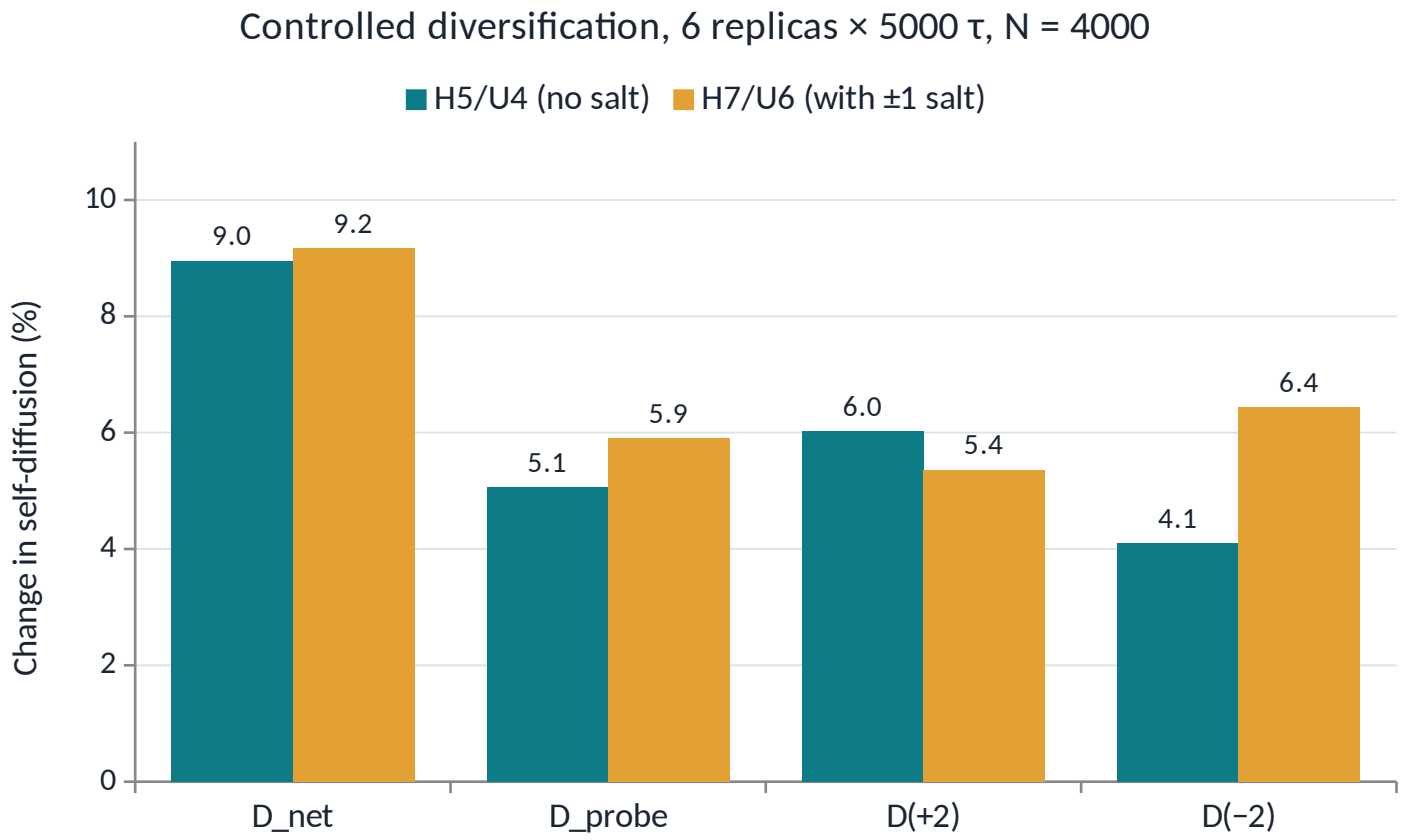
Full designs: Δs_ex ≈ +0.30 kB, probes +18–20%; each sign accelerates separately

Controlled diversity: Δs_ex = +0.103 kB, mean +8–9%, probes +5–6% (resolved by longer runs, next slide)

Labels only (L5/U4): identical diffusion in every replica

Salt only (B5/B3): Δs_ex slightly negative, no resolved speedup

Longer runs resolve a +5–6% ion response



Probe 95% CIs: H5/U4 [+4.0, +6.1]%, H7/U6 [+4.7, +7.1]%. H7/U6 is stable in every MSD window; H5/U4 drops to ≈ +3% beyond 100 τ, so it is quoted as +3 to +5%.

−7%

shear viscosity η (Green–Kubo), U4 → H5 and U6 → H7

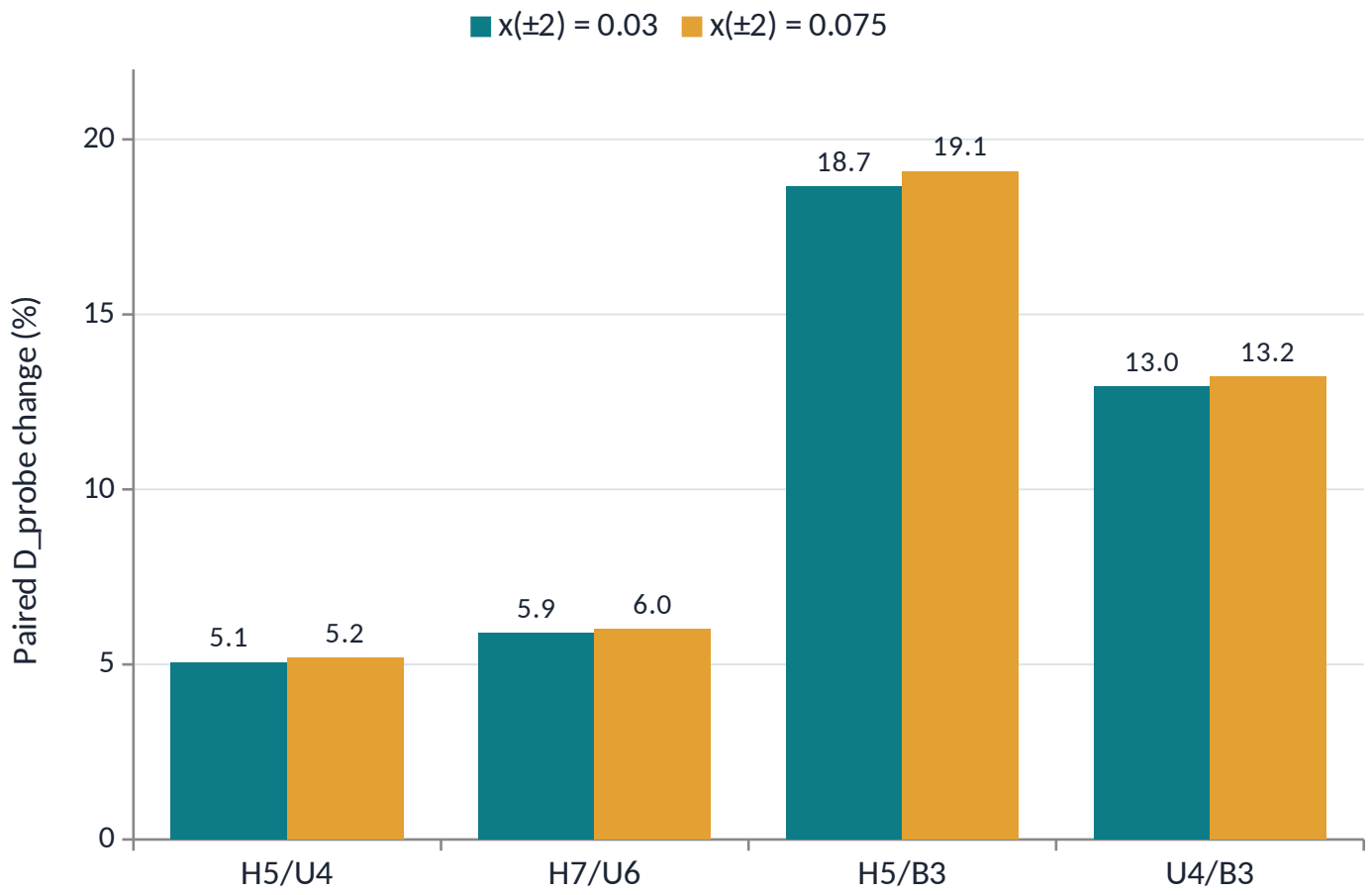
+1% vs −2 to −3%

D_net·η stays put (Stokes–Einstein); D_probe·η falls: probes only partly carried along

+5.3% / +6.0%

same probe contrasts at N = 8000; D shifts +0.8–1.3%, as Yeh–Hummer predicts

Tripling the divalent salt leaves the conclusions intact



+0.103 kB

Δs_ex(H5−U4) at both concentrations

−0.016 kB

s_ex shift as HNC predicted

0.73

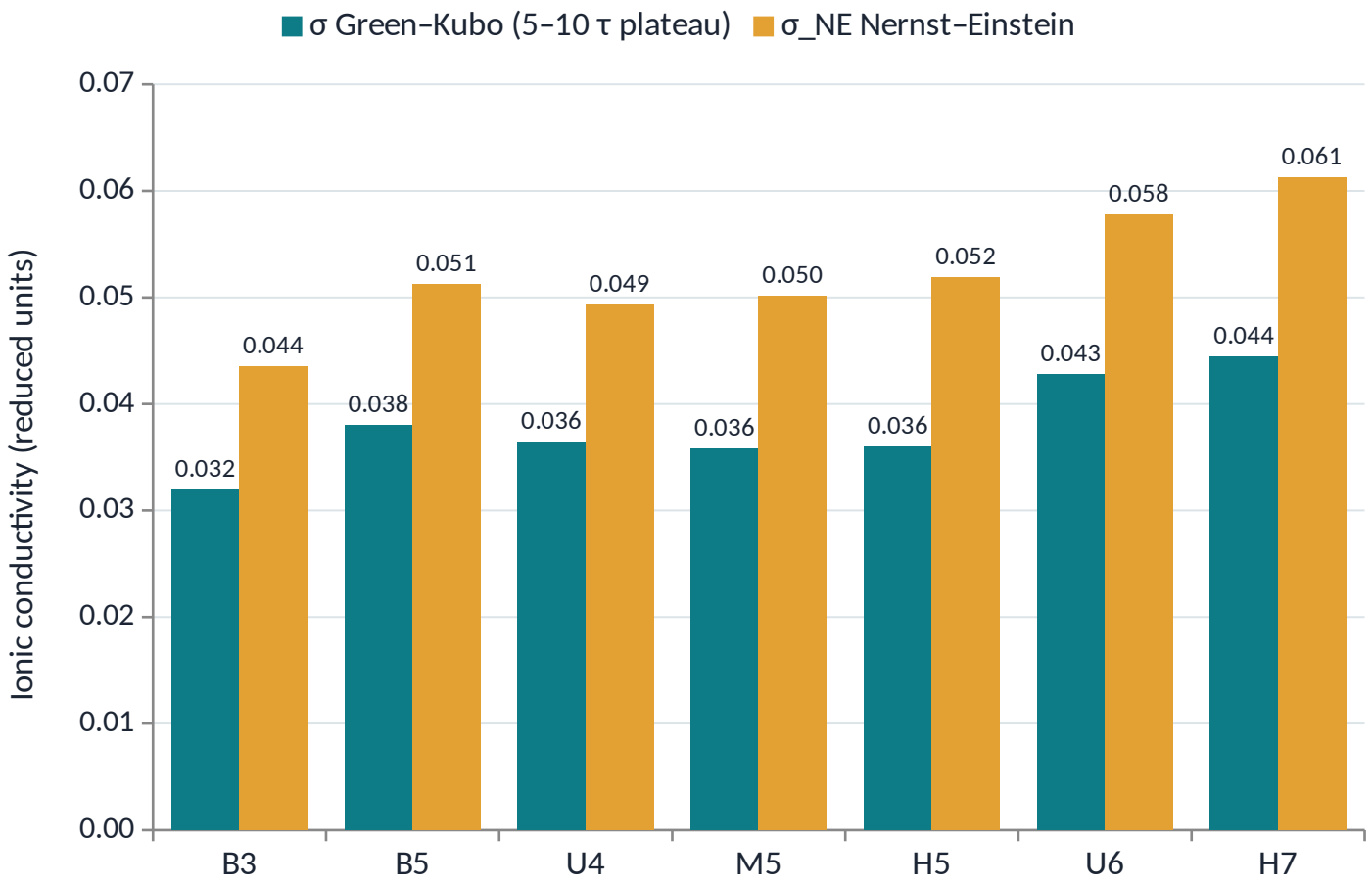
σ_GK / σ_NE in every state (95% CI 0.71–0.75)

η −9%

Diversification; σ unchanged within ±10%

x(+2)=x(−2)=0.075 (total 0.15), N=4000, 4 matched replicas × 3000τ, ρ=0.7, T*=2*

Conductivity follows carriers: σ ≈ 0.73 σ_NE



0.727

pooled σ/σ_NE, 28 replicas, CI 0.71–0.75; same in every state

+18.8%

B5/B3: extra ±1 carriers (NE predicts +17.7%)

+5–6%

diversification via NE; Green–Kubo change unresolved (±10%)

2–5 τ

Green–Kubo plateau; 2–20 τ cutoffs agree

$\sigma = (3Vk_BT)^{-1}\int\langle J(0)\cdot J(t)\rangle dt$ with $J = \Sigma\, q_i v_i$; $\sigma_NE = (\rho/k_BT)\, \Sigma\, x_i q_i^2 D_i$. $x(\pm 2) = 0.075$, 4 replicas × 3000 τ

High-divalent mixtures: state properties

State	s_ex/k_B	Z	u/ε	D_net	D_probe	η	σ	σ/σ_NE
B3	-1.987	2.503	-1.894	0.2204	0.2073	1.264	0.0320	0.734
B5	-1.990	2.497	-1.918	0.2197	0.2075	1.274	0.0380	0.741
U4	-1.788	2.753	-1.273	0.2583	0.2347	1.104	0.0365	0.740
M5	-1.758	2.665	-1.282	0.2647	0.2387	1.080	0.0358	0.714
H5	-1.685	2.463	-1.302	0.2807	0.2469	1.002	0.0360	0.694
U6	-1.793	2.747	-1.297	0.2578	0.2339	1.099	0.0428	0.741
H7	-1.688	2.457	-1.326	0.2812	0.2479	1.036	0.0444	0.725

−0.016 k_B

s_ex in every state vs x(±2) = 0.03 (HNC predicts the same)

−0.02 / −0.08

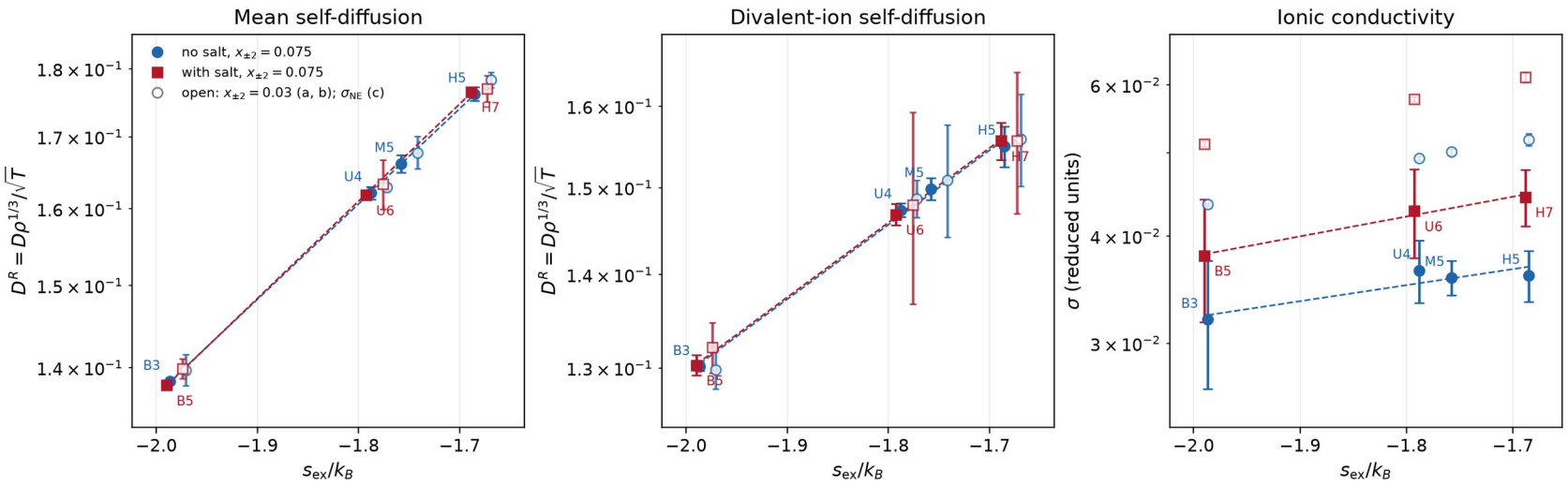
Z and u per particle: more Coulomb cohesion

−0.5 to −1.3%

D_probe; η unchanged within its CI: dynamics barely affected

Means over 4 replicas; D from the 20–100 τ MSD window, η and σ Green–Kubo. L5 ≡ U4 (label-only split).

TI entropy tracks mobility; HNC and s2 follow



Descriptive Rosenfeld fits (3–4 mixtures, $R^2 > 0.99$): B_net ≈ 0.8, B_probe ≈ 0.6; at x(±2) = 0.075: B_net 0.80–0.82, B_probe 0.59, local probe slopes 0.49–0.56 vs 0.81–0.84 for the mean. Absolute HNC entropies are 0.06–0.09 kB too low; increments are much better.

≤ 7%

HNC error on the controlled entropy increments

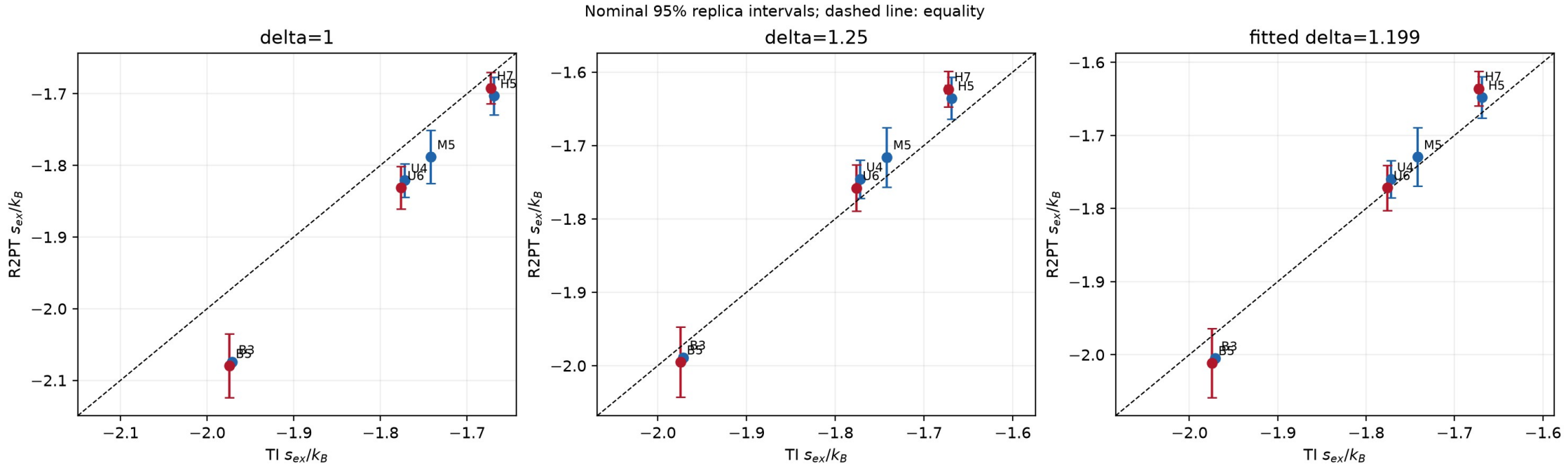
≈ 80%

share of s_ex carried by the MD pair entropy s_2, and of its increments

−0.33 kB

beyond-pair residual s_ex − s_2: composition dependent, not negligible

2PT when TI is unfeasible: calibrate, then rank



Calibrate one δ

Fit over 7 mixtures: $\delta^* = 1.20$ [1.17, 1.23]; $\delta = 1.25$ cuts RMS bias 0.067 → 0.029 kB; held-out bias 0.030 kB

Good within a family

M5/U4 and H5/U4 increments reproduced within 0.01 kB; label split exactly zero

Limits

H7/U6 +30%, baseline → design +17–23%; D enters the fluidicity, so 2PT cannot test an entropy–diffusion link on its own

Next: fast entropy estimators for electrolytes

2PT, velocity based

Ready

- Calibrated δ ≈ 1.2–1.25 for this model family
- One dense trajectory per candidate
- Use for ranking; recalibrate on TI anchors
- Not independent of diffusion

Voronoi cluster entropy

Open

- 24 trajectories packaged with TI, s_2 and D references
- Must reproduce TI order, the 0.03 and 0.10 kB increments and the label null
- Structure based, so independent of transport

1 HNC screen
candidate mixtures

2 TI anchors
2–3 states to calibrate and validate

3 2PT / Voronoi
rank many candidates

4 Conductivity
Green–Kubo current: σ ≈ 0.73 σ_NE at x(±2)=0.075

Next physics test: probe–solvent coordination, association lifetimes and exchange kinetics — does weaker local caging control the ion mobility?

Conclusions hold at x(±2)=0.075; conductivity is set by carrier number and D (σ ≈ 0.73 σ_NE), not by diversification